09. Certified Chern phases in a bounded tight-binding family

Pure Thought research portfolio

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

Two solids can both resist ordinary electrical conduction in their interiors yet differ in how their quantum states fit together. In a topological phase, that global organization cannot be smoothly changed without passing through a gap closing or leaving the assumptions that protect it. Here we build simple lattice models and prove which parameter ranges belong to which phase. The task is closer to drawing an exact map of a model than predicting a particular material.

Key terms

- **Hamiltonian H** — The mathematical operator specifying a quantum system's energy and dynamics.
- **Tight-binding model** — A lattice model in which particles occupy localized orbitals and move between them through specified hopping terms.
- **Orbital** — One of the local quantum states included at a lattice site or within a unit cell.
- **Band** — A range of allowed energies labeled by crystal momentum in a periodic system.
- **Band gap** — An energy separation between the selected occupied bands and the next bands.
- **Brillouin zone** — The space of distinct crystal momenta for a periodic lattice.
- **Chern number** — An integer measuring a global twist in the selected quantum states across a two-dimensional Brillouin zone.
- **Occupied projector** — An operator selecting the band states being treated as occupied.
- **Phase diagram** — A map showing which kind of behavior occurs for each model-parameter range.

Why this matters

Topological models explain why some properties depend on global structure rather than microscopic details. A certified phase map provides reliable examples for theory and simulation, and it helps identify which ingredients a proposed design actually needs. Proving a gap everywhere is important because a small missed gap closing can invalidate a computed topological label.

What progress would mean

A useful result would give explicit models, guaranteed parameter ranges and checked Chern numbers. These could guide later physical designs, while leaving material stability, fabrication and experimental behavior to additional work.

Core question

Within a declared finite-range Hamiltonian family, which parameter regions have a proven band gap and occupied-band Chern number, and what minimality statements follow within that family?

Scope and assumptions

Begin with the two-band square-lattice model $H_m(k) = \sin(k_x)\sigma_x + \sin(k_y)\sigma_y + [m + \cos(k_x) + \cos(k_y)]\sigma_z$, k∈[−π,π]^2. Use one occupied band and a fixed orientation. Analyze rational parameter intervals separated from m=−2,0,2.

For subsequent searches fix dimension, orbitals, filling, hopping support, coefficient domain, and symmetry representation. A rational grid is finite; a continuous coefficient box requires a different completeness argument. “All Hamiltonians for a space group” is not an enumerable specification.

Mathematical target

For a general family $H(k) = \sum_R t_R e^{ik\cdot R}$, impose t_−R=t_R† and the stated symmetry equations. Prove E_{r+1}(k)−E_r(k)≥δ>0 for all k and all parameters claimed. With P the occupied projector, choose

$$C = \frac{1}{2\pi i}\int_{BZ} \operatorname{Tr} P[\partial_x P, \partial_y P]\, d^2k.$$

Certify C through an analytic degree/homotopy argument or validated integration with an error enclosure containing a unique integer, including discretization and projector errors.

Required outputs and proof obligations

Return model coefficients, gap lower bounds, invariant and sign convention, certified parameter coverage, and uncovered boxes. A minimality claim must specify whether it minimizes orbitals, hopping range, or coefficients and must exclude every smaller object in that declared class.

Validation and rejection controls

Recover gap closings at −2,0,2 and trivial/nontrivial intervals on either side. Reverse orientation and verify C changes sign. A fine-grid lattice Chern calculation is a useful cross-check, but must not substitute for the continuous gap proof. Check a time-reversal-invariant occupied bundle has zero total Chern number.

Milestones and research extension

Baseline: complete phase diagram for H_m away from transition values. Strong: a certified atlas for one bounded symmetry-compatible family. Research extension: new range/band tradeoffs or minimality theorems, explicitly relative to the permitted family.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Chen et al., The impossibility of exactly flat non-trivial Chern bands in strictly local periodic tight binding models — A concrete example of why range/locality assumptions change realizability claims.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$H_m(k) = \sin k_x\, \sigma_x + \sin k_y\, \sigma_y + (m + \cos k_x + \cos k_y)\sigma_z.$$

$$E_{r+1}(k) - E_r(k) \geq \delta > 0, \qquad C = \frac{1}{2\pi i} \int_{\mathrm{BZ}} \mathrm{Tr}\Big(P[\partial_{k_x} P, \partial_{k_y} P]\Big)\; d^2k.$$

Prove the gap over the full claimed parameter and momentum domain. Fix orientation and the occupied projector. An analytic argument or validated integral enclosing one integer certifies C; a momentum mesh alone does not.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Major — unbounded classification (37–41, 803–813).** Fixing a space group does not bound the hopping range or continuous parameters. Claims to enumerate every model and prove minimality have no finite search universe.
2. **Blocking — false orbital lower bounds (811–813).** Higher Chern number does not generally require |C|+1 bands; two-band maps can have higher degree when suitable hopping harmonics are allowed. Range restrictions are indispensable to any proposed lower bound.
3. **Major — numerical integer versus certified topology (89–94, 790–793).** Discretized curvature and finite differences are not exact continuum certificates without gap and error control.

4. **Major — physical extrapolation (45–53).** A mathematical model does not imply a stable synthesizable material. The revision makes the model atlas the result and keeps materials relevance conditional.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **Within a declared finite-range Hamiltonian family, which parameter regions have a proven band gap and occupied-band Chern number, and what minimality statements follow within that family?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: complete phase diagram for H_m away from transition values. Strong: a certified atlas for one bounded symmetry-compatible family. Research extension: new range/band tradeoffs or minimality theorems, explicitly relative to the permitted family.

Sources supporting the corrections

- Chen et al., The impossibility of exactly flat non-trivial Chern bands in strictly local periodic tight binding models — A concrete example of why range/locality assumptions change realizability claims.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.

- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.